Self-Supervised Learning Machine for Electricity-Theft Detection to Secure Smart Grids

Sergey Gumirov
Scientific Supervisor: Elena Gryazina

Skolkovo Institute of Science and Technology

19.02.2026

Skoltech
Skolkovo Institute of Science and Technology

Outline

1. Related Work
2. Proposed Method: Dataset and Pipeline
3. Results
4. Conclusion
5. Future Work & Roadmap

Skoltech
Skolkovo Institute of Science and Technology

Motivation

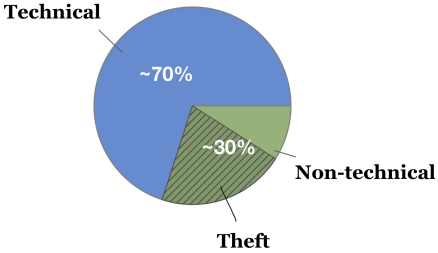


Fig. 1 - Electric network losses

- In many power systems, NTL is often reported at ∼**10–30%** of distributed energy (can reach ∼**40%** in high-loss regions).
- NTL is **mainly theft**: illegal connections, meter tampering; in this scheme theft is **60% of NTL**.
- **Impact:** worse load forecasting, biased state estimation/loss allocation and DER planning.
- **Moscow region example (2025):** reported damage **$3.6M/year**.

Skoltech
Skolkovo Institute of Science and Technology

Related Work: Supervised Approaches

Deep NN (LSTM / CNN)

Zheng et al., 2018

- Learns temporal usage patterns directly from load curves
- Captures nonlinear behavior without heavy manual rules
- Limitation: requires many reliable fraud labels

Wide & Deep / Hybrid Models

Jiang et al., Applied Energy, 2019

- Combines expert features with a deep branch
- Balances interpretable signals with learned representations
- Limitation: sensitive to class imbalance and noisy labels

Takeaway: strong predictive power, but label dependence remains the key bottleneck.

Skoltech
Skolkovo Institute of Science and Technology

Related Work: Unsupervised and Semi-Supervised

Isolation Forest

Liu et al., ICDM, 2008

- Isolates anomalies through random tree splits
- Scales well to large customer populations
- Limitation: strongly depends on feature design/preprocessing

Local Outlier Factor (LOF)

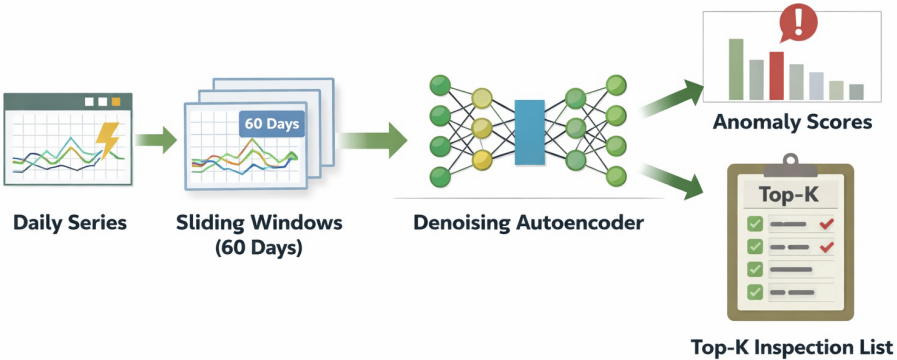
Breunig et al., SIGMOD, 2000

- Scores anomalies by local density differences
- Useful for local irregularities missed by global rules
- Limitation: sensitive to k, distance metric, and high dimension

Takeaway: label-free setup is attractive, but ranking quality depends heavily on representation choices.

Skoltech
Skolkovo Institute of Science and Technology

Proposed Method: Overview



Skoltech
Skolkovo Institute of Science and Technology

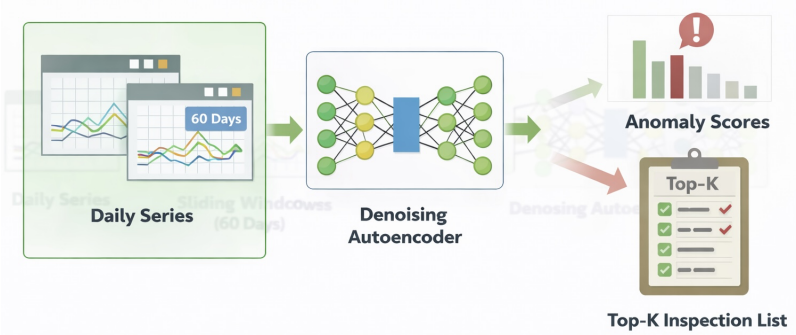
Dataset (SGCC): What It Contains

- SGCC smart meter dataset: electricity consumption time series per customer. **42,372 unique customers**
- Label `flag` $\in \{0, 1\}$ indicates confirmed/suspected theft cases. **3,615 thefts**
- Time span: **1034** days (after alignment/aggregation in our pipeline).
- Strong imbalance: theft share is **8.53%**.

Note: the exact counts below are reported after preprocessing (unique customers, daily aggregation).

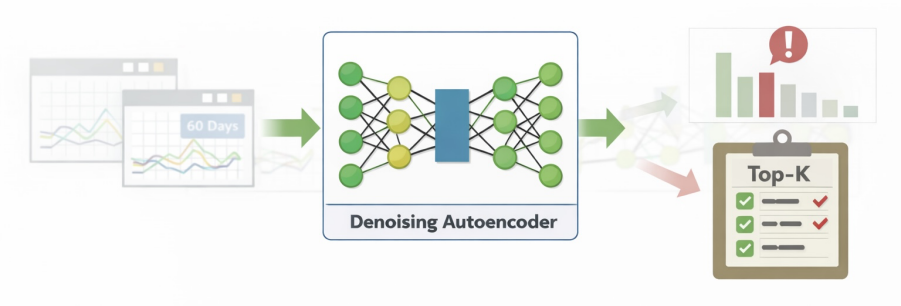
Skoltech
Skolkovo Institute of Science and Technology

Preprocessing Pipeline



1. Daily aggregation (sum consumption within each day).
2. Missing values: forward fill → backward fill → zero filling.
3. Per-customer standardization (z-score): remove scale, keep shape.
4. Sliding windows: length L days, stride S days.

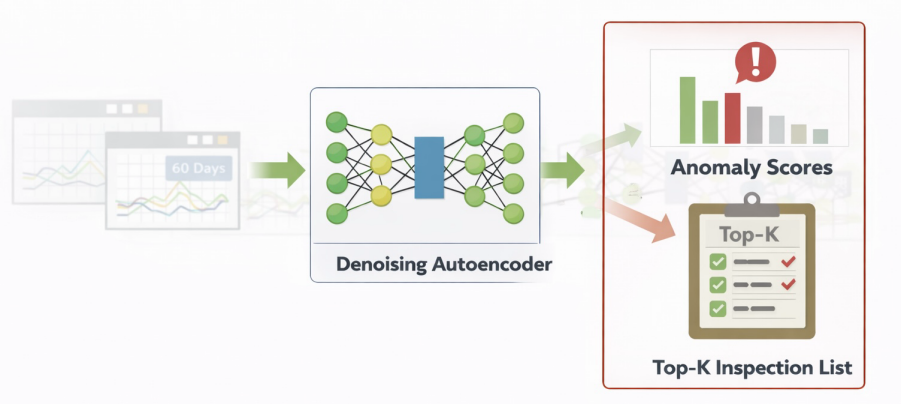
Proposed Method: Self-Supervised Denoising Autoencoder



- Input: standardized consumption window of length L.
- Add noise / dropout $\Rightarrow$ denoising objective prevents memorization.
- Train to minimize reconstruction MSE: $\mathrm{MSE}(x, \hat{x})$.

Skoltech
Skolkovo Institute of Science and Technology

Proposed Method: Top-K inspection list



- The 95th percentile of window reconstruction error is used as the anomaly score.
- Customers are ranked by this score.

Skoltech
Skolkovo Institute of Science and Technology

Customer Anomaly Score Distribution

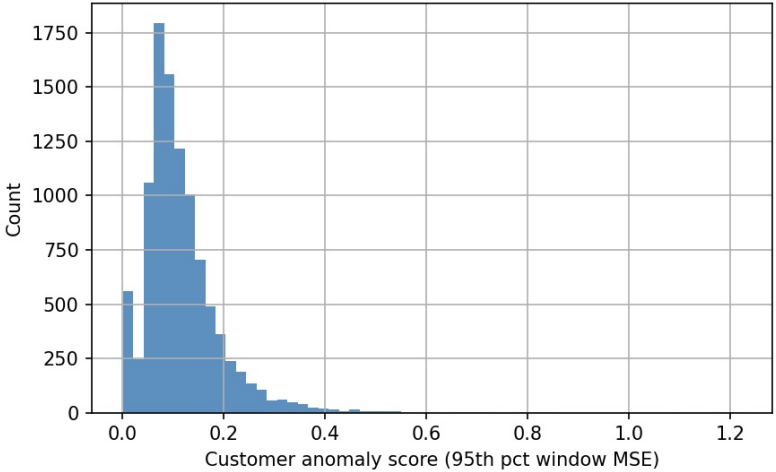


Fig. 2 - Distribution of customer-level anomaly scores

Skoltech
Skolkovo Institute of Science and Technology

Top-K Evaluation (Full Range)

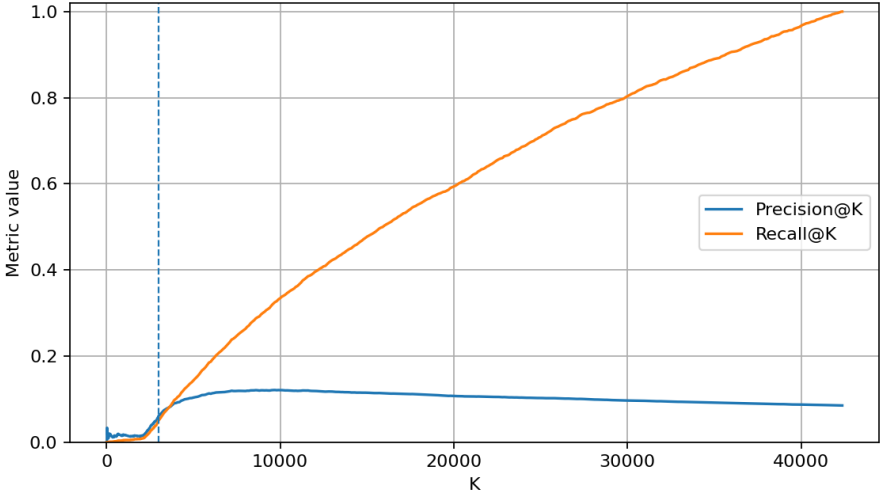
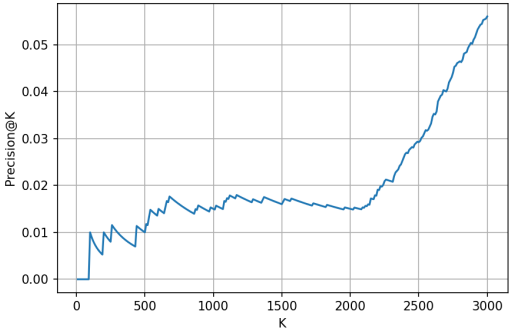


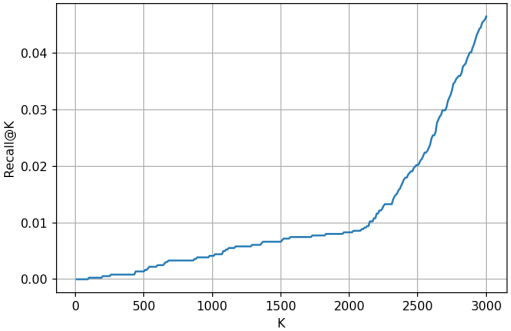
Fig. 3 - Precision@K and Recall@K across the full inspection range

Skoltech
Skolkovo Institute of Science and Technology

Top-K Evaluation (Operational Perspective)



(a) Precision@K vs number of inspected customers (K)



(b) Recall@K vs number of inspected customers (K)

Interpretation: K is chosen based on inspection budget and the precision/recall trade-off. We target **K = 3000** as a practical working point.

Skoltech
Skolkovo Institute of Science and Technology

Comparison with Related Work

Method	Learning Type	Metric	Value
Rule-based (utility reports)	Heuristic	Detection rate	$\sim$0.60
Isolation Forest (reported)	Unsupervised	ROC-AUC	0.75–0.90
Supervised LSTM (reported)	Supervised	F1-score	0.65–0.80
Proposed SSL Autoencoder	Self-supervised	ROC-AUC	**0.61**

Reported results are taken from literature using similar smart-meter datasets; direct comparison is limited due to differences in labeling and evaluation protocols.

Skoltech
Skolkovo Institute of Science and Technology

Conclusion

- Proposed a self-supervised pipeline for theft detection using a denoising autoencoder.
- Learned normal consumption patterns without requiring theft labels for training.
- Produced actionable Top-K ranked inspection lists consistent with operational constraints.
- Results provide a strong baseline and can be improved via semi-supervised calibration.

Skoltech
Skolkovo Institute of Science and Technology

Future Work (next iterations)

- **Stronger SSL**: contrastive SSL (TS2Vec/SimCLR-style for windows), masked reconstruction, forecasting pretext.
- **Better scoring**: robust aggregation (trimmed mean), change-point features, seasonal normalization.
- **Cost-aware ranking**: optimize for Precision@K / expected economic return (not only AUC).
- **Robustness**: concept drift checks (seasonality, tariff changes), outlier-resistant preprocessing.
- **Explainability**: per-window error attribution + prototype "similar normal windows".

Skoltech
Skolkovo Institute of Science and Technology

6-Month Plan (Jan–Jun)

Month	Milestones / Deliverables
Jan	Data audit: leakage checks; stronger missingness handling; reproduce current AE baseline end-to-end.
Feb	Benchmarking: supervised baselines (Wide&Deep CNN / simple LSTM) + Top-K metrics; define target K grid.
Mar	SSL upgrade v1: denoising + masked reconstruction; ablations (L, stride, noise); calibration of anomaly score.
Apr	SSL upgrade v2: contrastive / forecasting pretext; compare representations (UMAP/cluster stability); drift tests.
May	Cost-aware ranking: optimize Precision@K / utility value; error analysis by customer segments; interpretability report.
Jun	Packaging: clean training/eval scripts + report; draft paper/tech report; slides + demo notebook.

Skoltech
Skolkovo Institute of Science and Technology

Thank you for your attention!
Questions?

Skoltech
Skolkovo Institute of Science and Technology

References I